

Password

Importance

Passwords are an **integral** part of our digital lives, both at work and for personal use. They are the **key** to our digital home. A secure home cannot be opened with a master key.

Factors of Strength

The strength of a password is determined by its **complexity**: the stronger it is, the more resistant it will be face to cyber attacks (brute force attacks, dictionary attacks, etc.). It depends on three factors:

- **Length**: The number of characters in the password.
- **Alphabet**: The variety of characters used
- **Common Usage**: If common words are used.

Calculation of Strength

Number of Possibilities = Alphabet ^ Length of the Password

Example: Alphabet of 26 letters and length of 8 characters:

$$n = 26^8 = 2.08 \times 10^{11}$$

Entropy: $E = \log_2(n)$

Example: Alphabet of 70 symbols and 8 characters:

$$E = \log_2(70^8) = 49 \text{ bits (very low)}$$

- Minimum entropy recommended by ANSSI: **77 bits** (for a local usage).
- Strong password: minimum entropy turn around **100 bits**.

Protection in Databases

To secure passwords in databases, **hash functions** are used to avoid to storing them in plain. The checksum allow to verify the integrity of the data, so it can compare a data with the original, can be used to verify if the password is correct.

Hash functions are one-way, making it impossible to recover the original password from the hash. However, **rainbow tables** (pre-computed hashes) exist.

Websites like crackstation can sometimes recover passwords from classic hashes, so **salted hashes** are used. A salt who's unique per user is added to the original hash, significantly increasing security. This makes the hacker calculate 'x' hashes per user, instead of 'x' hashes total.

Conclusion

Password strength depends on complexity, length, and alphabet used. Entropy, calculated from the number of possible passwords, reflects the password's strength. Databases use hashing to protect passwords from being stored in plain and salting it to make the task more difficult for the hacker.